🐍 Seaborn Complete Notes for Beginners to Advanced

What is Seaborn?

Seaborn is a Python library used for **data visualization**. It is built on top of **Matplotlib** and provides a high-level interface for creating attractive and informative statistical graphics.

Released: 2014
Developed by: Michael Waskom

Why Use Seaborn?

- Beautiful graphs with minimal code
- Easy statistical visualization
- Works directly with Pandas DataFrames
- Built-in themes and color palettes
- Better default styles than Matplotlib
- Suitable for Data Analysis, Machine Learning, and Data Science

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Installation

```
pip install seaborn
```

Check version:

```python
import seaborn as sns

print(sns.__version__)
```

Import Libraries

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Built-in Datasets

```
tips = sns.load_dataset("tips")
iris = sns.load_dataset("iris")
titanic = sns.load_dataset("titanic")
penguins = sns.load_dataset("penguins")
diamonds = sns.load_dataset("diamonds")
flights = sns.load_dataset("flights")
```

Check dataset

```
tips.head()
tips.info()
tips.describe()
```

Common Plot Syntax

```
sns.plot_name(
    data=df,
    x="column1",
    y="column2"
)

plt.show()
```

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1. Scatter Plot

Shows the relationship between two numerical variables.

```
sns.scatterplot(
    data=tips,
    x="total_bill",
    y="tip"
)

plt.show()
```

Use Cases

- Age vs Salary
- Height vs Weight
- Experience vs Salary

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2. Line Plot

Shows trends over time or ordered data.

```
sns.lineplot(
    data=tips,
    x="size",
    y="tip"
)

plt.show()
```

Use Cases

- Stock Price
- Temperature
- Sales Trend

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3. Bar Plot

Displays average values for categories.

```
sns.barplot(
    data=tips,
    x="day",
    y="total_bill"
```

```
)

plt.show()
```

Use Cases

- Average Salary
- Average Marks
- Average Sales

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4. Count Plot

Counts the number of observations in each category.

```
sns.countplot(
    data=tips,
    x="day"
)

plt.show()
```

Use Cases

- Male vs Female
- Product Count
- Customer Categories

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5. Histogram

Shows the distribution of a numerical variable.

```
sns.histplot(
    data=tips,
    x="total_bill"
)

plt.show()
```

KDE (Kernel Density Estimation)

Creates a smooth probability density curve.

```
sns.histplot(
    data=tips,
    x="total_bill",
    kde=True
)

plt.show()
```

Why KDE?

- Smooth distribution
- Easy to identify peaks
- Better than only histogram bars

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6. Box Plot

Displays spread, median, quartiles, and outliers.

```
sns.boxplot(
    data=tips,
    x="day",
    y="total_bill"
)

plt.show()
```

Shows

- Minimum
- Q1
- Median
- Q3
- Maximum
- Outliers

Use Cases

- Salary Analysis

- Exam Marks
- Detect Outliers

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7. Violin Plot

Shows both box plot information and distribution.

```
sns.violinplot(
    data=tips,
    x="day",
    y="total_bill"
)

plt.show()
```

---

8. Strip Plot

Displays all individual observations.

```
sns.stripplot(
    data=tips,
    x="day",
    y="total_bill"
)

plt.show()
```

---

9. Swarm Plot

Avoids overlapping points.

```
sns.swarmplot(
    data=tips,
    x="day",
    y="total_bill"
)
```

```
plt.show()
```

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10. Pair Plot

Shows pairwise relationships among numerical columns.

```
sns.pairplot(tips)

plt.show()
```

Useful for Machine Learning and EDA.

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11. Heatmap

Visualizes a matrix using colors.

```
corr = tips.corr(numeric_only=True)

sns.heatmap(
    corr,
    annot=True,
    cmap="coolwarm"
)

plt.show()
```

Use Cases

- Correlation Matrix
- Feature Selection

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12. Correlation Matrix

```
tips.corr(numeric_only=True)
```

---

13. Hue Parameter

Adds different colors based on category.

```
sns.scatterplot(
    data=tips,
    x="total_bill",
    y="tip",
    hue="sex"
)

plt.show()
```

---

14. Style Parameter

Different marker styles.

```
sns.scatterplot(
    data=tips,
    x="total_bill",
    y="tip",
    style="smoker"
)

plt.show()
```

---

15. Size Parameter

Changes marker size.

```
sns.scatterplot(
    data=tips,
    x="total_bill",
    y="tip",
    size="size"
)
```

```
plt.show()
```

---

16. Palette

Color themes.

```
sns.barplot(
    data=tips,
    x="day",
    y="tip",
    palette="Set2"
)

plt.show()
```

Popular Palettes

- deep
- muted
- pastel
- bright
- dark
- colorblind
- viridis
- magma
- coolwarm

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17. Figure Size

```
plt.figure(figsize=(8,5))

sns.barplot(
    data=tips,
    x="day",
    y="tip"
)

plt.show()
```

18. Themes

```
sns.set_theme(style="darkgrid")
```

Styles

- darkgrid
- whitegrid
- dark
- white
- ticks

19. Titles and Labels

```
sns.barplot(data=tips, x="day", y="tip")

plt.title("Average Tips")

plt.xlabel("Day")

plt.ylabel("Tip")

plt.show()
```

20. Saving Graph

```
plt.savefig("graph.png", dpi=300)
```

Most Important Parameters

Parameter	Purpose
data	Dataset
x	X-axis column

Parameter	Purpose
y	Y-axis column
hue	Different colors
style	Marker style
size	Marker size
palette	Color theme
linewidth	Line thickness
alpha	Transparency
bins	Histogram intervals
kde	Density curve
annot	Show values in heatmap
cmap	Color map

Matplotlib vs Seaborn

Matplotlib	Seaborn
Basic plotting	Statistical plotting
More code	Less code
Manual styling	Automatic styling
Flexible	Easy to use
Base library	Built on Matplotlib

Advantages

- Easy syntax
- Beautiful graphs
- Statistical plots
- Built-in datasets
- Great with Pandas
- Excellent for Exploratory Data Analysis (EDA)

Disadvantages

- Less customization than Matplotlib
- Depends on Matplotlib
- Not suitable for highly customized graphics

Real-World Applications

- Data Analysis
- Machine Learning
- Business Intelligence
- Sales Dashboard
- Healthcare Analytics
- Finance
- Banking
- Marketing
- Customer Analytics

Interview Questions

1. What is Seaborn?
2. Why do we use Seaborn?
3. Difference between Matplotlib and Seaborn?
4. What is KDE?
5. What is a Heatmap?
6. What is Pair Plot?
7. What is Hue?
8. Difference between Strip Plot and Swarm Plot?
9. What is Box Plot used for?
10. What are built-in datasets in Seaborn?
11. What is `sns.load_dataset()`?
12. What is `palette`?
13. What is `annot=True`?
14. What is `cmap`?
15. Why is Seaborn preferred in Data Science?

Learning Roadmap

1. Scatter Plot
2. Line Plot

3. Bar Plot
4. Count Plot
5. Histogram
6. KDE
7. Box Plot
8. Violin Plot
9. Strip Plot
10. Swarm Plot
11. Pair Plot
12. Heatmap
13. Hue
14. Palette
15. Themes
16. Figure Size
17. Saving Figures

Summary

Seaborn is one of the most popular Python libraries for creating attractive statistical visualizations. It is built on Matplotlib, integrates seamlessly with Pandas, and is widely used in Exploratory Data Analysis (EDA), Machine Learning, and Business Analytics.